Home Page

This task provides you with experience in React and React Components. You are given the wireframe to expand the home page of your DEV@Deakin web platform.

Complete the following:

1. Complete the Task Requirements.
2. You will find "Topic Videos and Demo Videos" of Week 3 and Week 4 on the SIT313 CloudDeakin site particularly useful as a reference for this task.
3. Create a new GitLab repository for this task. Ensure all files/resources created are committed to this repository.
4. Create a 2 – 4 minute video where you provide a walkthrough of your code and solution, demonstrating your understanding of the concepts. Upload your video to Panopto.
5. Create a short report that includes:
    a. Screenshots of your DEV@Deakin web platform.
    b. A link to your task repository on GitLab.
    c. A link to your video.
6. **Upload** to OnTrack and **engage with feedback** to get this complete!

Task Requirements

1. GitLab repository

- In your GitLab account, create a New Project called Task P3. **Set the Visibility Level to *Internal*.** Failure to do this means you cannot have your submission marked as Complete.
- In your repository you **must** create a *.gitignore* file, and add inside it node_modules/ to prevent this folder from being tracked by Git.

Strictly do not include your node_modules folder in your GitLab repository. Failure to do so means you will receive a Resubmit for this task.

All code and resources (e.g. images) that you develop for this task must be committed to your Task P3 repository, including proper commit comments.

2. Home Page

Convert/migrate your DEV@Deakin web platform you have developed in HTML/CSS/JavaScript during the previous Pass tasks into a React project. You **do not** need to migrate your existing ExpressJS backend for this task – just migrate the frontend for now.

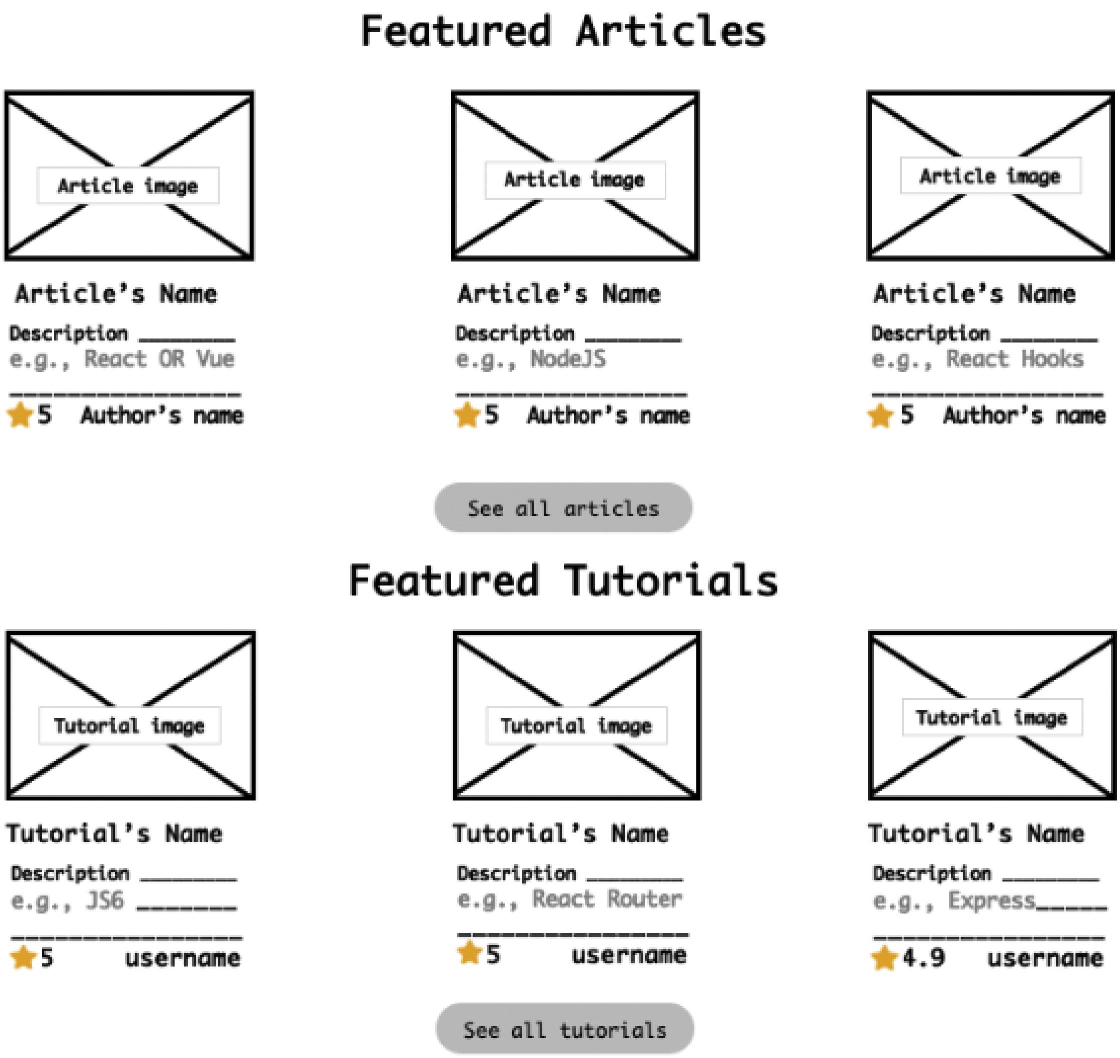
You can technically use the *create-react-app* package to create a React project, however it is outdated and no longer supported. See the **Optional Technologies** section for alternatives such as Vite.

Once you have migrated your DEV@Deakin web platform to a React project, add additional content to your home page that exactly match the Design Criteria listed below. Your solution must:

- Correctly utilise React components (in a parent->children) architecture. You cannot bundle all your code into one or two components.
- For the Articles, you must store these in an array object and use the map() function.
- Incorporate media such as images and icons to enhance the look and feel of your page. There are many free image libraries such as Pexels and Pixabay, and free icon libraries such as Lucide.

3. Design Criteria

Review the following design criteria and add to your existing home page – your solution must match *exactly* the same formatting. However, you are free to incorporate your own designs by choosing your own styles (e.g. colours, shading, fonts, etc) and adding additional features/components/media as you wish.



Add a Footer to your page, in line with the following example:

Optional Technologies

We strongly encourage you to explore additional technologies that are often standard in industry, that may assist you in completing this task:

- TypeScript – Adds 'type safety' to JavaScript. Standard JavaScript is not type-safe, meaning you can store any datatype in any variable which often leads to bugs. Strongly recommended to explore, either within this task or in your own time, and is expected to use within industry.
- Vite – A modern, faster alternative to the now out-dated Create React App. It is a lightweight frontend framework, usable with React.
- NextJS – A more powerful, but a more structured framework that combines both frontend and backend into the one package. A slightly steeper learning curve, but enables much more control over page rendering.
- Tailwind CSS: A CSS framework where you combine your CSS code with the HTML code. Allows you to design your pages faster.
- CSS Flexbox – A layout model for arranging your HTML elements in a much more flexible, dynamic way. Here is a great guide.
- CSS Grid – A layout model for arranging your HTML elements in a grid-layout. Here is a great guide.
- UI component library – These come with many pre-made elements (buttons, sliders, tables) to speed-up development time, and to create a more consistent style across your website. Some recommended ones include PrimeReact, Shadcn, Mantine, and many more.

Optional VS Code Extensions

VS Code has a gigantic library of extensions that can improve and speedup the development process. Here are some that may assist you:

- Tailwind CSS IntelliSense by Tailwind Labs – If using Tailwind CSS, this is a must-have extension to enable auto-complete when typing.
- Prettier by prettier.io – Automatically formats your codes' style and structure, to enforce a consistent style throughout.

Available Resources

Feeling a bit lost?

No problem at all! Your tutors are experts and would love to assist. Ensure to attend the (you can attend as many seminars as you like) and speak with a tutor, or write a message on the SIT313 Teams channel.

We have many resources to assist in your learning journey – if you are ever unsure then please reach out to us, we are always happy to help.

- **The SIT313 Teams channel** – an active learning resource with weekly announcements from the Unit Chair, and regular discussions/assistance from the SIT313 teaching team and students.
- **Seminars** – you can attend both on-campus and online sessions, and join as many as you like each week. Your tutors will provide demonstrations and assistance to help guide your learning.
- **SIT313 CloudDeakin** – contains a large collection of text and video resources.
- **Announcements** - Keep an eye on your email and any announcements made in the SIT313 Teams channel.

Caution: Acceptable use of Artificial Intelligence

It is acceptable to use AI as a learning tool. However, you can **never** ask AI to write code to complete some/parts of this task for you: this is cheating because it does not demonstrate that you understand anything. Detection of inappropriate use of

> AI has a very real consequences, such as failing the task, or even failing the entire unit.